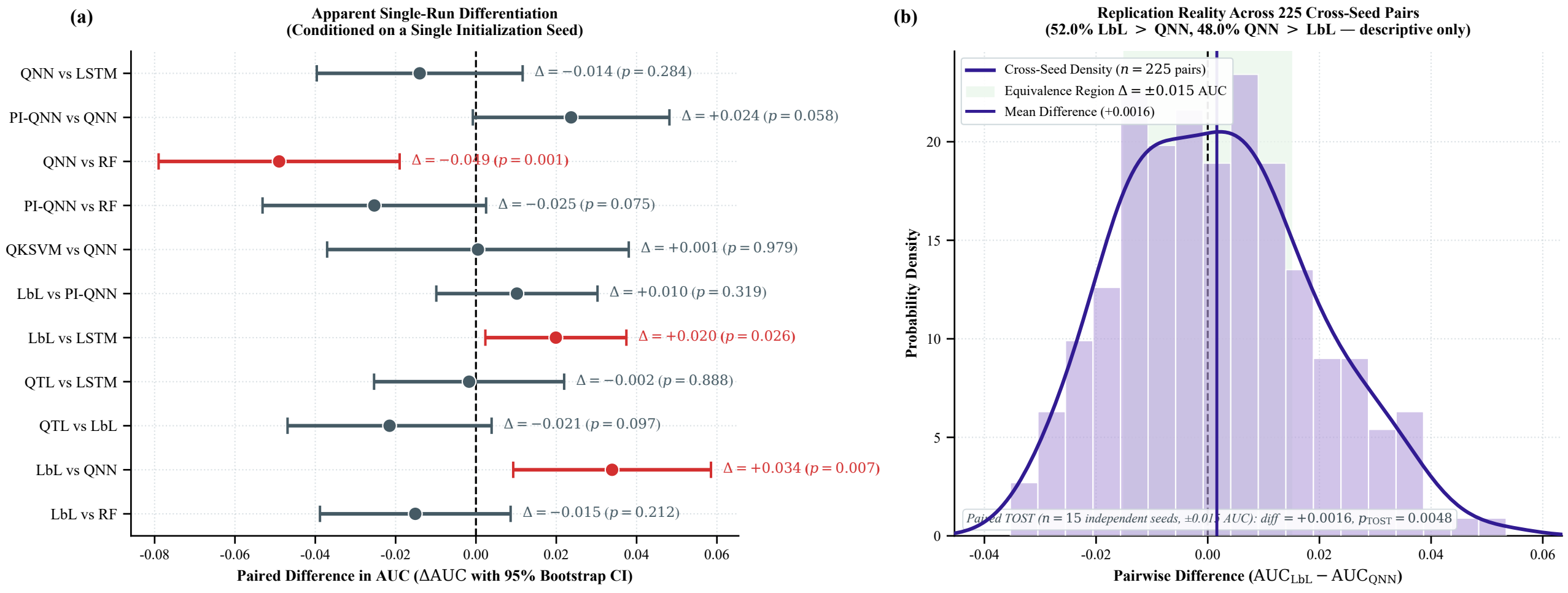

(a)
Apparent Single-Run Differentiation
(Conditioned on a Single Initialization Seed)
QNN vs LSTM
PI-QNN vs QNN
QNN vs RF
PI-QNN vs RF
QKSVM vs QNN
LbL vs PI-QNN
LbL vs LSTM
QTL vs LSTM
QTL vs LbL
LbL vs QNN
LbL vs RF
Δ = −0.014 (p = 0.284)
Δ = +0.024 (p = 0.058)
Δ = −0.049 (p = 0.001)
Δ = −0.025 (p = 0.075)
Δ = +0.001 (p = 0.979)
Δ = +0.010 (p = 0.319)
Δ = +0.020 (p = 0.026)
Δ = −0.002 (p = 0.888)
Δ = −0.021 (p = 0.097)
Δ = +0.034 (p = 0.007)
Δ = −0.015 (p = 0.212)
-0.08
-0.06
-0.04
-0.02
0.00
0.02
0.04
0.06
Paired Difference in AUC (ΔAUC with 95% Bootstrap CI)
(b)
Replication Reality Across 225 Cross-Seed Pairs
(52.0% LbL > QNN, 48.0% QNN > LbL — descriptive only)
Cross-Seed Density (n = 225 pairs)
Equivalence Region Δ = ±0.015 AUC
Mean Difference (+0.0016)
Probability Density
0
5
10
15
20
Paired TOST (n = 15 independent seeds, ±0.015 AUC): diff = +0.0016, pTOST = 0.0048
-0.04
-0.02
0.00
0.02
0.04
0.06
Pairwise Difference (AUCLbL − AUCQNN)